The International Journal of Advanced Manufacturing Technology (2024) 134:5463–5479
https://doi.org/10.1007/s00170-024-14368-z

ORIGINAL ARTICLE

Research on pipeline intelligent welding based on combined line structured lights vision sensing: a partitioned time–frequency-space image processing algorithm

Zhu Chuanhui[1,2] · Wang Zihao[1,2] · Zhu Zhiming[1,2] · Guo Jichang[3]

Received: 27 March 2024 / Accepted: 1 September 2024 / Published online: 20 September 2024

Abstract
By applying vision sensing to realize online intelligent control on the pipeline welding process, the welding efficiency and qualification rate of pipeline girth butt joint during on-site construction of long-distance oil and gas pipelines can be effectively improved. During the pipeline external welding process, based on the designed combined line structured lights vision sensor (C-LSLVS), a partitioned time–frequency-space (PTFS) image processing algorithm is proposed to extract the laser centerline for the deformed laser lines image of double-V composite groove. The algorithm first utilizes information in the time domain to remove random interference from the image, and then, based on Radon transform (RT) and discrete Fourier transform (DFT), other stationary interferences can be eliminated and laser line characteristics can be enhanced by back-projection reconstruction in the frequency domain and space domain, so the stable and accurate extraction of laser centerline can be achieved under different situations. Subsequently, the theoretical analysis and error calculation of the pipeline groove sizes and related parameters solution model based on the local plane fitting method are carried out, and high-precision calculations of the parameters are completed. Finally, an intelligent pipeline welding system based on the C-LSLVS is constructed. In the intelligent welding experiment, the maximum detection error of the pipeline welding groove sizes did not exceed 0.20 mm, and the weld seam tracking deviation did not exceed 0.25 mm. The welding results show that the system has important application value for the development of pipeline intelligent welding.

Keywords Vision sensing · Image processing · Time–frequency-space · Parameters calculation · Model analysis · Pipeline intelligent welding

1 Introduction

Long-distance oil and gas transport pipelines, serving as lifeline of energy supply, have become an essential energy infrastructure related to national energy security, social and economic development, and people's well-being [1–4]. During their on-site laying construction, the pipeline girth butt joint with pre-machined welding groove is preheated firstly, then completed by inside root welding using an internal welding machine, and outside hot welding, filling and covering welding using an external welding machine successively. The external welding machine commonly faces numerous unfavorable factors that affect the stability of the welding process and the adaptability of the equipment when implementing the external welding processes on construction sites, and operators still need to monitor the welding state, and manually fine-tuning the process parameters at all times to control the external welding machine. Due to the fact that operators rely mainly on visual perception for their adjusting actions, adopting online visual feedback control based on visual information is a significant direction for further advancement in the welding field [5–7].

Compared with passive vision sensing, active vision sensing, especially line structured light vision sensor (LSLVS), has advantages of higher detection accuracy, convenient operation and lower cost, etc. [8, 9]. It has been widely used

✉ Zhu Zhiming
zzmdme@tsinghua.edu.cn

1 Department of Mechanical Engineering, Tsinghua University, Beijing 100084, China

2 Key Laboratory for Advanced Materials Processing Technology, Ministry of Education, Beijing, China

3 Coal Mining Branch, China Coal Research Institute, Beijing 100013, China

in the welding field, such as welding groove sizes detection, weld seam tracking, and welding torch spatial position and posture (SPP) parameters detection and control [10–13].

The foundation of the application of LSLVS in intelligent welding is the accurate and stable online identification of the imaging characteristics of deformed laser lines projected onto the welding groove (weld seam). There are two main forms of pipeline welding grooves: single V-groove and double-V composite groove (the typical cross-sectional dimensions of the two types of grooves are shown in Fig. 1). Many researchers [14, 15] have carried out research on the image processing algorithms for the extraction of the imaging features of the deformed laser line projected onto the single V-groove by line structured light. Gu et al. [16] realized laser centerline extraction through region of interest (ROI) segmentation, median filtering, thresholding, denoising, thinning, and curve fitting. By using the traditional morphological method and the spatiotemporal context (STC) tracking algorithm, Zou et al. [17] extracted the seam feature points in deformed laser line images with strong interference of V-shaped groove. Zhang et al. [18] designed two different image processing algorithms based on the Canny operator and skeleton thinning to extract the feature points of deformed laser lines image of single V-groove. Currently, with the continuous increase of pressure and flow of long-distance oil and gas pipelines, the steel materials grade, pipe diameter, and wall thickness are constantly increased [19]. To reduce the amount of weld metal filling and improve the welding efficiency, the double-V composite groove (Fig. 1(b)) with a small cross-sectional area is the preferred choice. The characteristics of deformed laser line image of the double-V composite groove are significantly different from the single V-groove (Fig. 1(a)) due to its special structure during the production process. It is difficult to accurately extract image features of deformed laser lines of the double-V composite groove by using the abovementioned single V-groove image processing flow and algorithms. In this situation, Zhu et al. [20] proposed a partitioned laser line image feature extraction method. Through image partitioning and corresponding processing design, the extraction of multi-features image information can be achieved. However, during the pipeline welding production process with numerous interferences, this method has poor anti-interference and adaptability. The laser line image processing method based on deep learning effectively improves the anti-interference, adaptability, and success rate of image processing under numerous interferences, but it requires a large amount of image data acquisition for model training [21–24]. Therefore, there is an urgent need for an image processing algorithm to realize stable and accurate laser lines extraction of the deformed laser lines image of the double-V composite groove in pipeline welding production.

The realization of intelligent welding based on LSLVS requires not only real-time and accurate feature information extraction from the deformed laser line images, but also real-time solution of the feature parameters such as welding groove sizes and relative SPP parameters of welding torch according to the image feature information. Researchers [25, 26] generally solve the parameters directly based on the deformed laser line feature points obtained from image processing, which shortens the accuracy and robustness of parameters solution. To improve them, Zhu et al. [27] proposed a novel approach to solve the parameters, which made full use of the deformed laser line data in the image by using the data segmentation and plane fitting. Furthermore, high-precision integrated detection of single V-groove sizes and welding torch relative SPP parameters was achieved. Shao et al. [28] used line structured lights vision sensing and directly employed plane fitting instead of the local curved surface for parameter solution of the curved workpieces, but did not clarify the theoretical basis for the applicability of this approach. When solving the normal vector of the local curved surface using line structured light features, Xue et al. [29] directly approximated

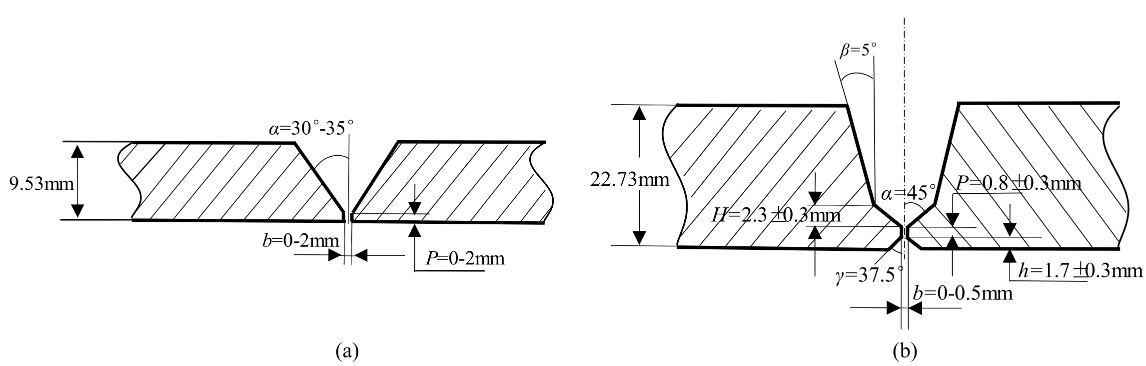


Fig. 1 The cross-sectional dimensions of pipeline girth butt joint welding grooves: (**a**) Single V-shaped welding groove; (**b**) Double-V composite welding groove

the local curved surface as a plane, and also did not give the numerical basis for the workpiece surface with a small curvature to be approximated as a plane. When using planes instead of local curved surfaces to solve parameters such as groove sizes, there will inevitably be some theoretical errors under different application conditions. It is necessary to analyze its theoretical errors in a targeted way to provide theoretical support and applicability guidance for its application.

When applying LSLVS to the pipeline external welding process, it is necessary to achieve stable and real-time extraction of laser line information in the image with interferences, as well as accurate calculation of feature parameters. Only then can we achieve accurate control of welding torch during the welding process, thereby obtaining a stable welding process and high-quality weld formation.

This paper studies the main issues in the implementation process of intelligent pipeline external welding based on the combined line structured lights vision sensor (C-LSLVS). The second part introduces the composition, composition, detection principle, and application of the sensor. In the third section, for the deformed laser line images collected by the sensor during pipeline welding process, a partitioned time–frequency-space (PTFS) image processing algorithm is proposed. In the fourth section, the pipeline groove sizes and related parameters are calculated by the local plane fitting method, and the theoretical analysis and error calculation are carried out; finally, a pipeline intelligent welding system based on the C-LSLVS is constructed, and some experiments were carried out.

2 Composition and application characteristics of the vision sensor

2.1 Structure design and detection principle of the vision sensor

The designed vision sensor mainly consists of an industrial camera, camera lens, two line laser emitters, and narrow-band filter, as shown in the dashed box in Fig. 2. During design, the central axes of the two line laser emitters are set parallel to each other and coplanar with the optical axis of the camera, and the designed angle between the central axes and the optical axis is 30°. The primary component and corresponding parameters of the C-LSLVS are shown in Table 1 [27].

According to the four rectangular coordinate systems (the image coordinate system: O-xy, the pixel coordinate system: o-uv, the camera coordinate system: O_C-$X_CY_CZ_C$, and the world coordinate system: O_W-$X_WY_WZ_W$) established in Fig. 2, a detection mathematical model of the C-LSLVS can be established as formula (1). It can realize the conversion from the pixel coordinates (u, v) of an image point p to the corresponding 3D coordinates (X_C, Y_C, Z_C) of the sampled point P in the O_C-$X_CY_CZ_C$ [30].

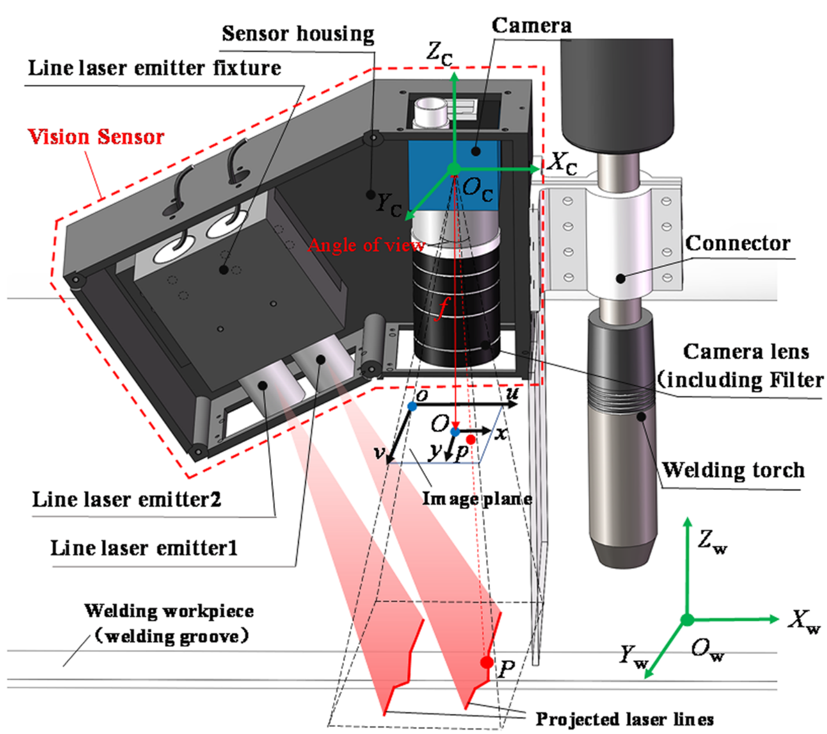


Fig. 2 Composition and detection principle of the C-LSLVS

Table 1 The primary component and corresponding parameters of the C-LSLVS

Primary components	Model and main parameters
Industrial CMOS camera	Model, MER2-503-23GM Resolution, 2448 × 2048 Exposure mode and frequency, global shutter, 23.5 fps Pixel size, 3.45 μm × 3.45 μm
Line laser emitter	Model, EL650-200G18LP Wavelength, 650 nm Lens, glass lens
Camera lens	Model, M1228-MPW3 Focal length, 12 mm Angle of view (D × H × V), 49.3° × 40.3° × 30.8° Working distance, 100 mm ~ inf
Filter	Filter wavelength, 650 ± 5 nm

$$\begin{cases} Z_C\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} X_C \\ Y_C \\ Z_C \\ 1 \end{bmatrix} \\ A_iX_C + B_iY_C + C_iZ_C + D_i = 0\,;\; i = 1,2 \end{cases} \tag{1}$$

In formula (1), u_0, v_0, f_x, and f_y are camera internal parameters. A_i, B_i, C_i, and D_i are the equation parameters of the structured light planes projected by the line laser emitters i (i = 1, 2). These two types of parameters are collectively called the vision sensor internal parameters.

Therefore, we can obtain the 3D point cloud coordinates of the laser lines projected onto the welding workpiece, and then the feature parameters can be calculated for feedback control.

2.2 Application of the vision sensor in pipeline all-position welding

As shown in Fig. 3, the pipeline all-position intelligent welding system is composed of the pipeline welding robot and the designed vision sensor. The pipeline welding robot runs on a rigid track and has five degrees of freedom, namely walking freedom D_v, horizontal lateral freedom D_e, height adjustment freedom D_h, tilt angle freedom D_α, and swing angle freedom D_β. The designed C-LSLVS is installed at the end of the welding actuator and located on the forward direction side of welding torch, which can detect the position of the welding torch and thus provide a detection basis for adjusting the position and posture of welding torch. With the movement of the trolley, the vision sensor can online detect and calculate the welding groove sizes and the deviations of the welding torch relative to the welding position so that the weld seam tracking and the action adjustment of the welding torch can be achieved through real-time feedback control during the pipeline all-position welding process, that is, the pipeline all-position intelligent welding is realized.

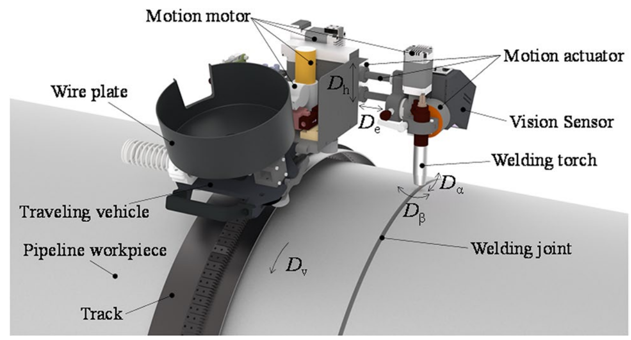


Fig. 3 Composition of pipeline all-position intelligent welding system

3 Deformed laser lines image processing of double-V composite groove

The key of automatic welding torch control and intelligent welding is stable and accurate feature extraction from deformed laser lines image. Compared with the single V-groove, the difficulty to extract the features of deformed laser lines image when LSLVS is applied to the double-V composite groove mainly lies in (1) regional characteristic differences: the narrow width and large depth of the groove, combined with the smaller slope angle of the upper V-groove surface, which lead to significant differences in laser line characteristics such as grayscale value at different areas and difficult to extract feature information from the entire image; (2) multiple interferences: the grooves with large depth are prone to contain more random interference such as spatter and intense arc light during the external welding process; besides, it is often necessary to polish the end surface and bottom groove surface of the pipeline before the external welding, which cause a significant amount of reflection interference in the collected laser line images. Figure 4 shows the deformed laser lines image of double-V composite groove captured by the C-LSLVS during welding process. Aiming at the problems of unclear and uneven laser line features, random interference, and polished reflective interference in the image, this paper proposes a PTFS image processing algorithm to extract laser centerlines of the deformed laser lines image during the welding process.

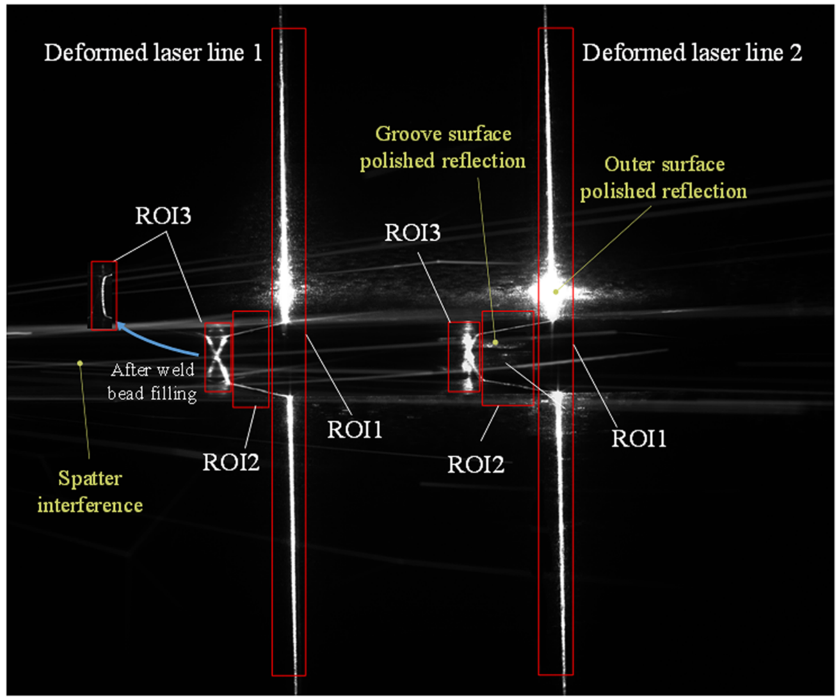


Fig. 4 Deformed laser lines image characteristics and sub-regions division of double-V composite groove

3.1 Sub-regions division of the deformed laser lines image

Take a single deformed laser line in the image as an example. Considering that the image characteristics of different regions are obviously different, it can be divided into three sub-regions of interest, as shown in Fig. 4, namely the vertical laser line region (ROI1), the upper V-groove laser line region (ROI2), and the lower deformed laser line region (ROI3). After filling welding, the weld bead makes the projected laser line imaging in ROI3 approximately deformed into a quadratic curve. There are significant differences of the image characteristics in the three sub-regions, so they are independently segmented and feature extracted in parallel to enhance the real-time performance and accuracy of the overall image processing.

To obtain the positions of three sub-regions in the entire image in real time, based on the vertical grayscale accumulation feature value of the preprocessed image as formula (2), the column position of ROI1 and ROI3 where the vertical laser line and the lower deformed laser line are located can be obtained, as shown in the col$i$ ($i=1, 2$) value in Fig. 5(a). The range between the columns of the two regions represents the column width and position of the ROI2 where the upper V-groove laser line is located.

$$\mathrm{col}(j) = \sum_{i} g(i,j) \tag{2}$$

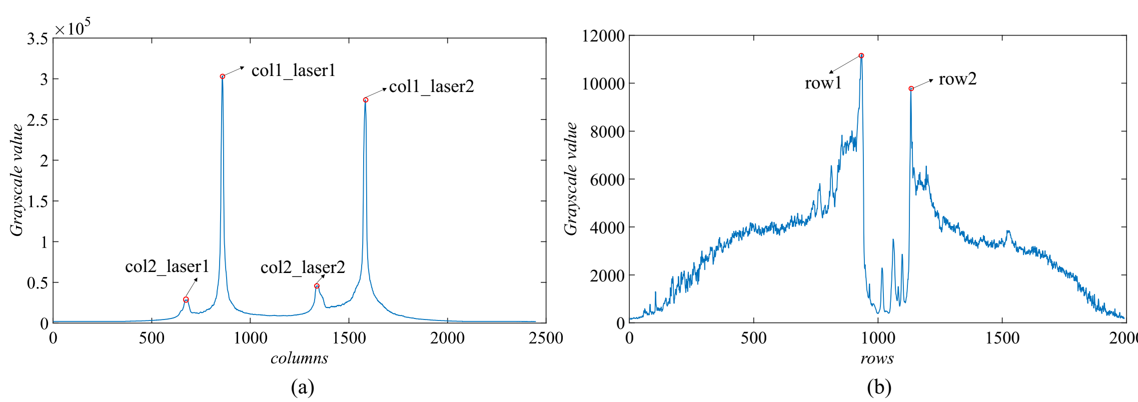


Fig. 5 Regional grayscale value accumulation: (**a**) Preprocessed image; (**b**) ROI1 (Deformed laser line 1)

In formula (2), $g(i, j)$ is the grayscale value of the pixel with coordinates $[i, j]$ in the preprocessed image.

Then, by accumulating the horizontal grayscale feature value of the ROI1 where the vertical laser line is located, we can obtain the two gray value sudden drop points in the horizontal grayscale accumulation feature value as shown in the row$i$ ($i = 1, 2$) value in Fig. 5(b). The range between the rows of the two points represents the row height and position of the ROI2. Similarly, the position and size of the ROI3 can be obtained through the same process and the prior shape of the deformed laser line.

3.2 Image processing of sub-regions

After completing the division of the three sub-regions, their image processing can be carried out simultaneously.

A PTFS image processing algorithm is proposed to extract the laser centerlines for deformed laser line images with complex characteristics and numerous interferences. The process of this algorithm is shown in Fig. 6.

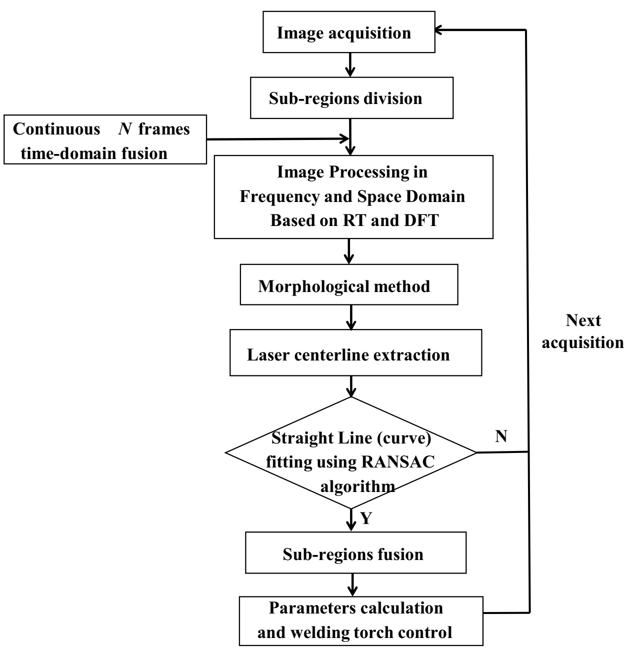


Fig. 6 A PTFS image processing algorithm flowchart

3.2.1 Time-domain fusion

During the welding process, random interferences such as arc light and spatters appear irregularly in each frame of the image, which are uneven and time-varying. To eliminate the random interferences, time-domain fusion processing can be performed on adjacent N frames of images. Within several sampling cycles, due to the small movement amplitude of the welding torch relative to the workpiece, there will not be significant differences in the position and shape of the laser line in adjacent N frames of the image. Therefore, the minimum gray value of the corresponding pixel points in adjacent N frames of the image can be taken to form a new image, which is used to eliminate random interferences, that is [31]:

$$I_{1,t}(x,y) = \min\{I_{0,t}(x,y), I_{0,t-1}(x,y), \cdots, I_{0,t-(N-1)}(x,y)\} \quad (3)$$

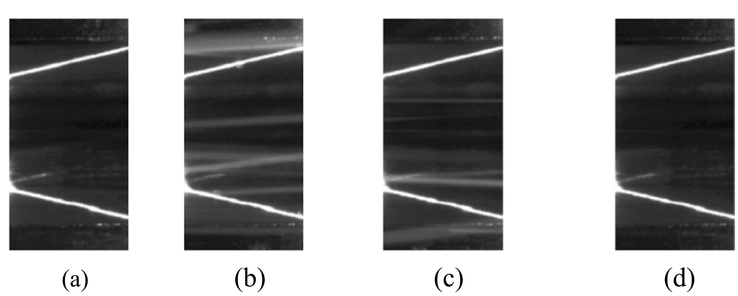
In formula (3), $I_{0,t}(x,y)$ is the image captured by the camera at time t, and $I_{1,t}(x,y)$ is the new image obtained by fusing N frames of images.

Since it will be relatively time-consuming to fuse the entire image, this step uses segmented sub-regions for fusion. This paper takes $N = 3$ in formula (3) to eliminate random interference. Taking ROI2 as an example, it can be seen that ROI2 in all three frames in Fig. 7(a)–(c) has varying degrees of spatter. After time-domain fusion, the random interference can be greatly eliminated, as shown in Fig. 7(d).

3.2.2 Image processing in frequency and space domain based on RT and DFT

The image processed by time-domain fusion can only eliminate some random interferences, but some non-time-varying interferences such as polished surface reflection cannot be eliminated. At the same time, the laser line information in the sub-regions is unclear, and extracting laser centerlines directly has poor stability and anti-interference properties. To these problems, based on the Radon transform (RT) and discrete Fourier transform (DFT), this section comprehensively utilizes the frequency domain and space domain information in the image to achieve the elimination of large-scale reflective interference, and enhances the laser line information in the region by back-projection reconstruction. On

Fig. 7 Time-domain fusion processing of sub-region images: (**a**) ROI2 at time t; (**b**) ROI2 at time t-1; (**c**) ROI2 at time t-2; (**d**) ROI2 after time domain fusion



this basis, the stable and accurate laser centerline extraction is achieved. The workflow of this algorithm is shown in Algorithm 1.

Similarly, taking the processing of the ROI2 sub-region after time-domain fusion as an example. Since the laser lines in each sub-region have a single direction feature, the RT can be used to extract this feature.

The RT is an integral transform, which is to perform a line integral of a function $f(x, y)$ defined on a 2D plane along a straight line in any direction θ on the plane, which can be expressed as [32]:

$$R(\theta, \rho) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\delta(x\cos\theta + y\sin\theta - \rho)dxdy \tag{4}$$

In formula (4), δ is the Dirac function; $x\cos\theta + y\sin\theta = \rho$ indicates that the line integral is performed along this straight line, where θ is the projection angle and ρ is the projection distance; $R(\theta, \rho)$ is the line integral value at θ and ρ.

The elimination of the interference requires filtering processing, and filtering processing is typically performed in the frequency domain, so it is necessary to perform DFT on the function $R(\theta, \rho)$ of the integral image. The Fourier transform (FT) of the projection of ρ is:

$$G(\omega, \theta) = \int_{-\infty}^{\infty} R(\theta, \rho)e^{-j2\pi\omega\rho}d\rho \tag{5}$$

In formula (5), ω is the frequency variable.

Algorithm 1 Frequency space domain image processing algorithm (FSDIPA)

1: **procedure** FSDIPA(*image*)	Image processing domain
2: *projection_image* ← performing RT on *image*	Space domain
3: *DFT_image* ← performing DFT on *projection_image*	Frequency domain
4: Apply filters for filtering the image	Frequency domain
5: Performing Inverse Discrete Fourier Transform (IDFT) on filtered	Frequency domain
6: *projection_image* ← setting the projection angle line integral values of the corresponding area to zero.	Space domain
7: *image* ← back-projection and interpolation processing on *projection_image*	Space domain
8: **return** *image*	
9: **end procedure**	

According to Fourier's slice theorem, the FT of a projection is equal to a slice of the 2D FT of the original function, which can be expressed as formula (6). Therefore, performing frequency domain transformation and subsequent processing on the function $R(\theta, \rho)$ is equivalent to performing corresponding operations on the original function $f(x, y)$.

$$G(\omega, \theta) = F(u, v) = [\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x, y)e^{-j2\pi(ux/M+vy/N)}]_{u=\omega\cos\theta; v=\omega\sin\theta} \tag{6}$$

The above formula (6) is the DFT of function $f(x, y)$, and (M, N) represents the size of image.

For non-time-varying interference such as polished surface reflection in the image, since it has specific frequency information, corresponding filters can be designed for filtering. The surface reflection interference in the sub-region is mainly low-frequency noise, so the Butterworth high-pass filter is used to eliminate the interference in the section, as shown in formula (7).

$$H_B(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}} \tag{7}$$

In formula (7), taking the cutoff frequency $D_0 = 10$ and filter order $n = 2$. The ROI2 images before and after filtering are shown in Fig. 8, the figure also shows the comparison with the processing results in the space domain.

Figure 8(a) is the original image with polished surface reflection; Fig. 8(b) and (c) show the images directly segmented with different methods in the space domain, where we can see that the direct segmentation method cannot reasonably eliminate interference and preserve laser lines information, and further extraction of laser line information is difficult; Fig. 8(d) is the image after Butterworth high-pass filtering (the image is displayed in the spatial domain), and Fig. 8(e) is the image of Fig. 8(d) after morphological processing. It can be seen in Fig. 8(d) and Fig. 8(e) that large-area reflective interference has been effectively removed.

At the same time, before performing the inverse transformation on $F(u, v)$, windowed filtering is also necessary to eliminate the blurring in the back-projection image, and Shepp-Logan filters can achieve smooth constraints on function, which can be expressed as [33]:

$$H_{S-L}(\rho) = \begin{cases} |\rho| \sin c(\omega/2B), |\omega| < B \\ 0 \quad , |\omega| \geq B \end{cases} \tag{8}$$

In formula (8), $B = 1/2d$, and d is maximum cutoff frequency.

Thus, the combined filter in the frequency domain can be represented as formula (9):

$$H_F = H_B \bullet H_{S-L} \tag{9}$$

After performing IDFT on the filtered image, the processed projection image can be obtained, which is:

$$R_1(\theta, \rho) = \frac{1}{MN} \sum_{x=0}^{M-1} F(u, v) H_F e^{j2\pi(ux/M+vy/N)} \tag{10}$$

In formula (10), $R_1(\theta, \rho)$ is projection function after filtering.

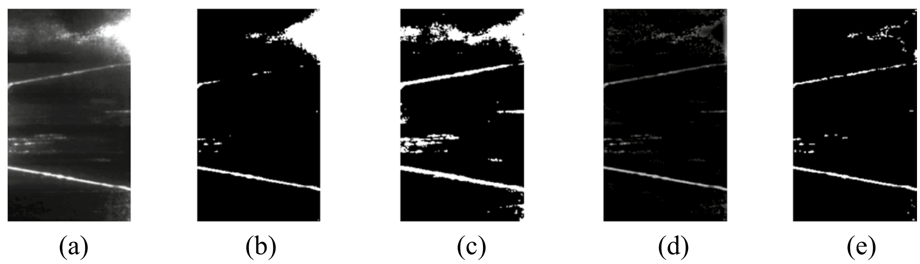


Fig. 8 ROI2 with polished surface reflection and corresponding processing: (**a**) Original image; (**b**) OTSU segmentation; (**c**) Adapative threshold segmentaion; (**d**) High-pass filtered imge; (**e**) Morphological processing for filtered imge

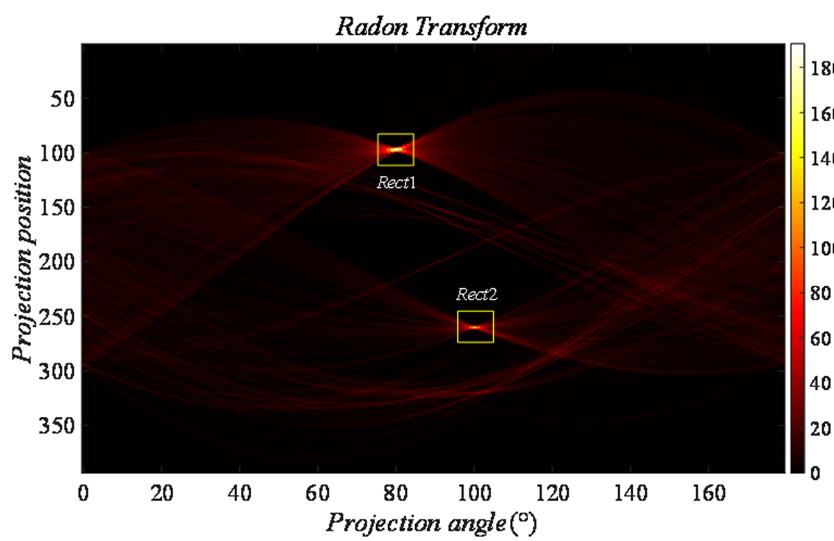


Fig. 9 Projected integral image of Radon transform

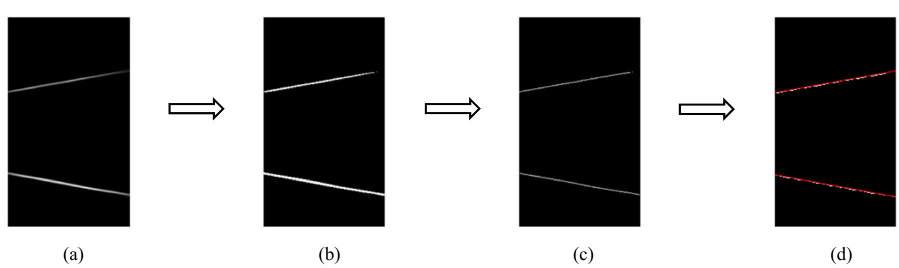


Fig. 10 Reconstructed laser line image and post-processing: (**a**) Reconstructed image of R_2; (**b**) Morphological processing; (**c**) Stripe center extraction; (**d**) Straight line fitting

According to formula (4) of RT, if there is a straight line in the image, the straight line will correspond to the maximum value in $R_1(\theta, \rho)$, and the image of $R_1(\theta, \rho)$ is shown in Fig. 9.

For the projected integral image in Fig. 9, we can find the two maximum value points within the two sets of prior angle ranges, then retain the values near the two points (as shown in the *Rect*1 and *Rect*2 in Fig. 9), and set the values at other positions to zero, that is formula (11).

$$R_2(\theta, \rho) = \begin{cases} R_1(\theta, \rho), & (\theta, \rho) \in Rect1URect2 \\ 0 & others \end{cases} \quad (11)$$

Then, the reconstructed laser line image can be obtained by back-projection and interpolation to the processed projection image at each angle, which is shown in Fig. 10(a).

As can be seen from Fig. 10, the interference information is almost completely eliminated, and the laser line information is enhanced and clear. After morphological processing and stripe center extraction using the Zhang-Suen skeleton thinning algorithm [34], the straight line equation of the laser centerline can be obtained by the RANSAC method, which can be expressed as formula (12).

$$\rho_{ui} = x\cos\theta_{ui} + y\sin\theta_{ui,} i = 1, 2 \quad (12)$$

In formula (12), ρ_{ui} and θ_{ui} represent the equation parameters of the corresponding straight lines obtained in ROI2.

Similarly, the same processing method is adopted for ROI1 and ROI3. The image processing process of the two sub-regions is shown in Fig. 11.

In Fig. 11, it can be seen that after frequency domain filtering, the large-area polished surface reflection interference in ROI1 (Fig. 11(a)) and ROI3 (Fig. 11(b)) can be eliminated, and the laser line information can be clearly reconstructed and enhanced through Radon transform domain image processing and back-projection. After weld bead filling (Fig. 11(c)), since there is no polished reflection interference in ROI3 and the laser line imaging information is clear, threshold segmentation, morphological processing, centerline extraction, and curve fitting can be directly performed.

3.3 Sub-regions refusion

After sub-regions segmentation and extraction of fitting laser centerline are completed, the image information of the three sub-regions can be fused according to the expression in formula (13).

$$\begin{cases} f_j(u_i, v_i) = 0, (u_i, v_i) \in P_j \\ [u_i, v_i] = R_j[u, v], j = 1, 2, 3 \end{cases} \quad (13)$$

In formula (13), (u_i,v_i) represents the sub-pixel coordinates of points on fitting laser centerline in the corresponding sub-region coordinate systems; (u,v) represents the sub-pixel coordinates of points on fitting laser centerline after fusion in the entire image coordinate system; R_j represents the transform matrix between different sub-region coordinate systems and the entire image coordinate system; $f_j(u_i,v_i)$ represents the implicit expression of fitting straight line or curve equations in different sub-regions, and P_j represents the point set on fitting laser centerline in different sub-regions.

Taking a deformed laser line as an example, its sub-region fusion diagram is shown in Fig. 12. Through the fusion of multiple sub-regions by formula (13), the laser centerline and its fitting straight line (curve) equation of the deformed laser line in the entire image can be obtained, thus laying the foundation for subsequent high-precision parameters calculation and motion control of the welding torch.

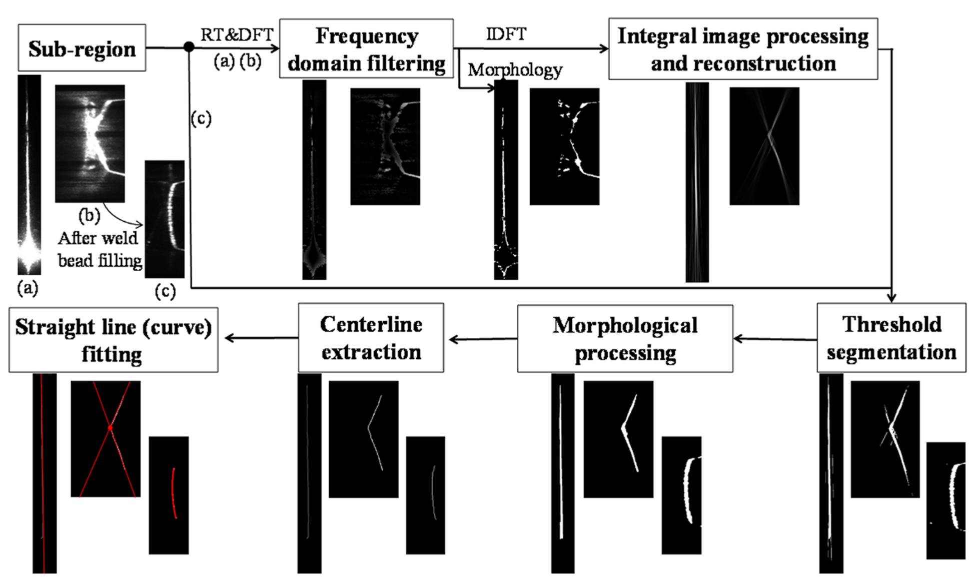


Fig. 11 Sub-regions of ROI1 and ROI3 image processing process. (**a**)–(**c**): (**a**) the upper part of ROI1 (the processing of the lower part is the same as upper part), (**b**) ROI3 without weld bead filling, (**c**) ROI3 after weld bead filling

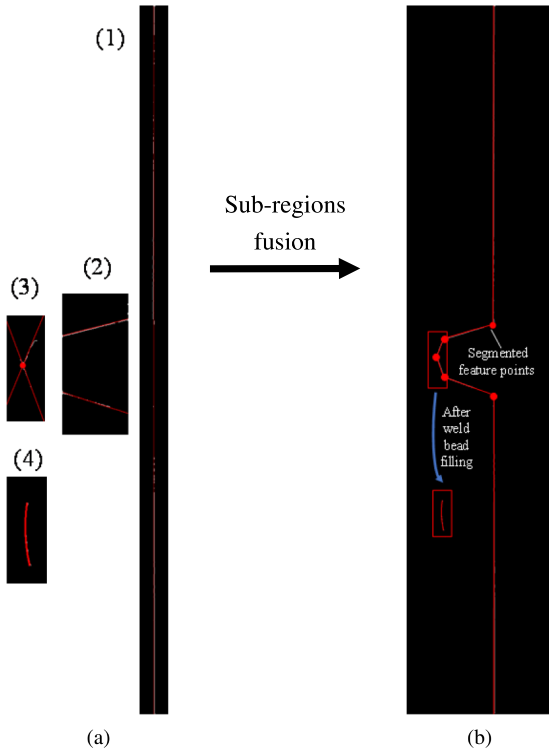


Fig. 12 Fusion of sub-regions after image processing. (**a**) Sub-regions after image processing, (1)–(4) in (a): (1) ROI1, (2) ROI2, (3) ROI3 without weld bead filling, (4) ROI3 after weld bead filling; (**b**) Fused deformed laser centerline and its fitting line

4 Solution and analysis model of pipeline groove sizes based on plane fitting

4.1 Calculation of pipeline groove sizes based on plane fitting

After obtaining the accurate laser centerline equation in the deformed laser line image, the method based on plane fitting can achieve the reconstruction of local groove surfaces and the calculation of feature parameters. Firstly, the laser line data is segmented according to the intersection points of fitting laser centerline equations, and then the segmented laser line data is mapped into a 3D point cloud according to the detection model formula (1). By performing plane fitting on the segmented 3D point cloud, the local groove surfaces can be reconstructed and their plane equations can be obtained. Therefore, the groove size parameters can be obtained, including groove width, depth, and surface angle. At the same time, the SPP parameters of the welding torch relative to the groove can also be calculated, such as relative position deviation and relative attitude angle [27]. The process is shown in Fig. 13.

Compared with the method of groove sizes calculation based on feature points of deformed laser lines [25, 26], this parameters calculation method based on the plane fitting makes full use of the deformed laser line data in the image, so the related size parameters of groove can be calculated with high accuracy and sound robustness.

4.2 Analysis of the calculation model using the plane fitting instead of the local curved surface

Since the pipe outer surfaces and its groove surfaces are all revolution surfaces (cylindrical or conical), thus, the method of using plane fitting instead of the local curved surface to solve parameters such as groove sizes must have an inevitable theoretical error, and the magnitude of this theoretical error will directly affect the detection accuracy of pipeline groove sizes. Therefore, it is necessary to analyze the model theoretical error under different conditions so as to provide credible support for the application of this method.

This section only analyzes the theoretical error for the solution of the groove sizes induced by using the plane fitting method instead of the local pipe outer surfaces and its groove surfaces, but does not consider the impact on groove sizes calculation error caused by other factors, such as pixel deviation in image feature extraction and machining deviation of the groove sizes. According to the installation position of the vision sensor on the end of the actuator of the pipeline welding equipment, the established pipe coordinate system O_P-$X_PY_PZ_P$, the camera coordinate system O_C-$X_CY_CZ_C$, the cross-sectional dimensions of the pipeline double-V composite groove, and the relative relationship between them are illustrated in Fig. 14.

In Fig. 14, the double-V composite groove of the pipeline is a symmetrical groove; points A, B, C, D, and E are the feature points of the cross-section of the double-V composite groove; R_p is the radius of the pipe outer surfaces; e is the difference between the origin O_P and the groove center point C along the Y_P axis in the pipe coordinate system; B, H, H_d, β, and α are groove size parameters.

Considering that there may be deviations in both the installation of the vision sensor and the track on the pipeline, the relationship between the two coordinate systems can be expressed as formula (14):

$$\begin{bmatrix} X_C \\ Y_C \\ Z_C \\ 1 \end{bmatrix} = \mathrm{R}_X\mathrm{R}_Z \begin{bmatrix} X_P \\ Y_P \\ Z_P \\ 1 \end{bmatrix} + {}^{C}\mathrm{p}_P : \mathrm{R}_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta_X) & -\sin(\theta_X) & 0 \\ 0 & \sin(\theta_X) & \cos(\theta_X) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \mathrm{R}_Z = \begin{bmatrix} \cos(\alpha_Z) & -\sin(\alpha_Z) & 0 & 0 \\ \sin(\alpha_Z) & \cos(\alpha_Z) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^{C}\mathrm{p}_P = [\begin{matrix} X_0 \\ 0 \\ Z_0 \\ 0 \end{matrix}] \tag{14}$$

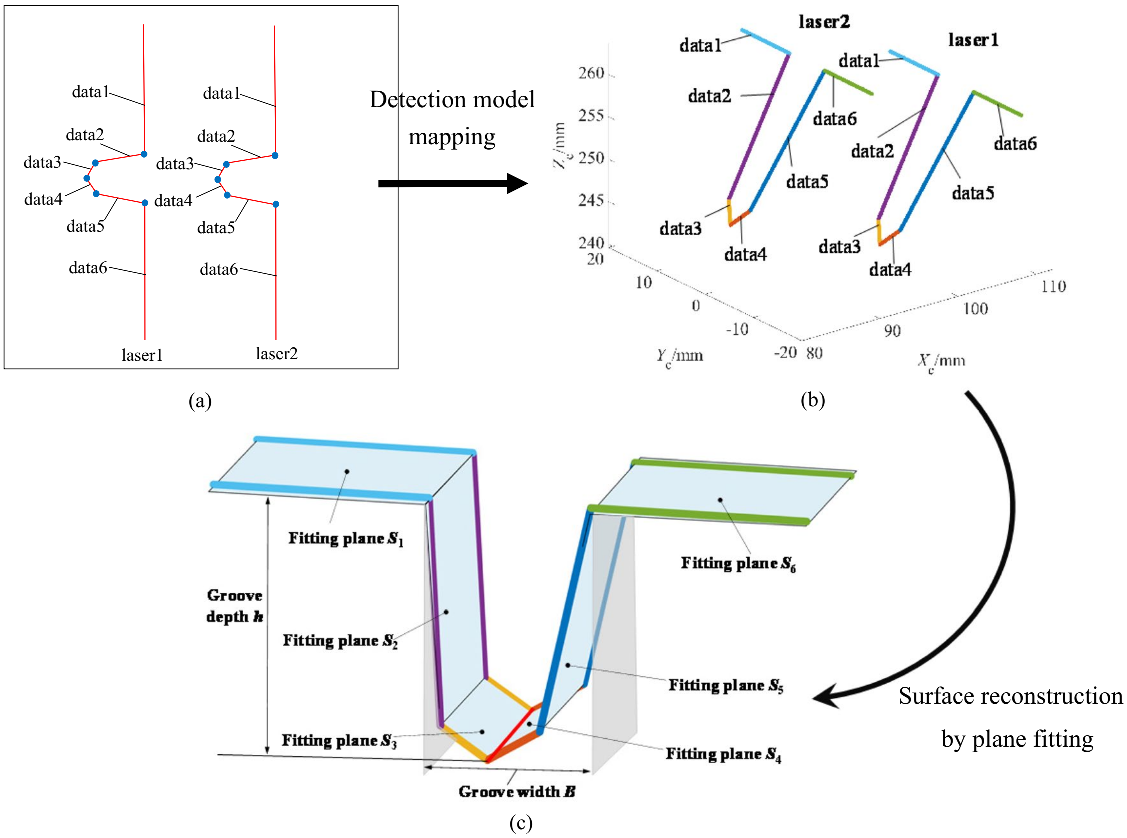


Fig. 13 The process of solving welding groove parameters based on plane fitting: (**a**) Data segmentation of fitting laser centerline; (**b**) 3D segmented data of laser lines; (**c**) Reconstruction of welding groove surfaces based on plane fitting

In formula (14), R_Z is the rotation matrix of the O_P-$X_PY_PZ_P$ around the Z_C axis of the O_C-$X_CY_CZ_C$ by the deviation angle α_Z; R_X is the rotation matrix obtained by the new coordinate system around the new X_C axis through the deviation angle θ_X, and ${}^{C}p_P$ is the position coordinates $(X_0, 0, Z_0)$ of the origin O_P in the O_C-$X_CY_CZ_C$.

The laser projection plane equation in the O_C-$X_CY_CZ_C$ in formula (1) is transformed into the plane equation in the O_P-$X_PY_PZ_P$ according to the coordinate transformation formula (14), as shown in formula (15).

$$A_pX_C + B_pY_C + C_pZ_C + D_p = 0 \tag{15}$$

In formula (15), $A_p = A_i\cos\alpha_Z + B_i\cos\alpha_Z + C_i\sin\alpha_Z\sin\theta_X$, $B_p = -A_i\sin\alpha_Z + B_i\cos\alpha_Z\cos\theta_X + C_i\sin\theta_X\cos\alpha_Z$, $C_p = -B_i\sin\theta_X + C_i\cos\theta_X$, and $D_p = A_iX_0 + C_iZ_0 + D_i\ (i = 1, 2)$ represent the corresponding laser plane equation parameters.

According to the dimensions of the cross-section in Fig. 14, the rotating surface equations of each outer surfaces of the pipeline can be calculated in the O_P-$X_PY_PZ_P$. When combined with the laser projection plane equations formula (15), the segmented expression of the deformed laser intersection lines can be obtained. The obtained local rotating outer surfaces and segmented intersection lines are shown in Fig. 15(a). Then, the obtained 3D segmented laser lines on different local outer surfaces are fitted with planes, respectively, and the reconstructed 3D local groove features are shown in Fig. 15(b). Based on this, the feature parameters under plane fitting model can be obtained. Compared

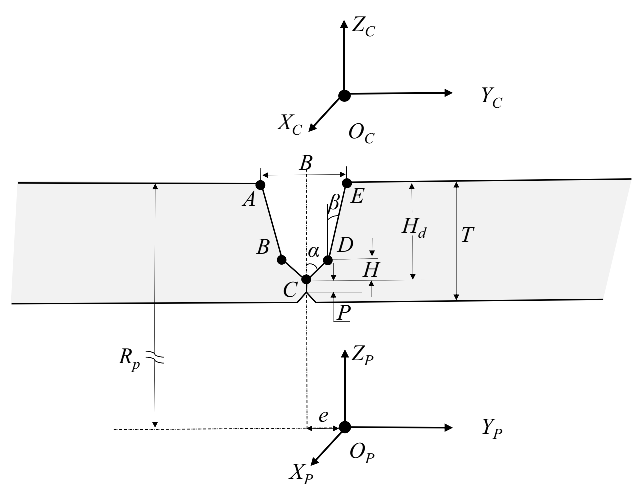


Fig. 14 The feature points of the double-V composite groove and coordinate system relative relationship

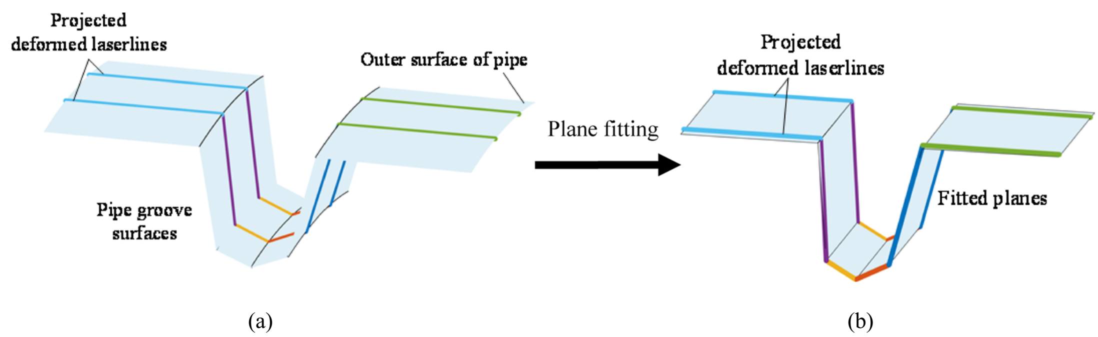


Fig. 15 The model schematic diagrams of plane fitting instead of the local curved surface: (**a**) Local pipe outer surfaces and its groove surfaces and 3D deformed laser lines; (**b**) Local plane fitting for segmented 3D deformed laser lines

Table 2 Typical parameter settings for the theoretical model

Parameter name	Parameter symbol	Parameter value
Pipe diameter	D_p	813 mm
Groove depth	H_d	20.2 mm
Groove width	B	8.10 mm
Pair gap	b	0
Groove surface angle	α/β	45°/5°
Lower groove height	H	2.5 mm
Weldment thickness	T	22.73 mm
Blunt edge	P	0.8 mm
Projected light plane equation parameters of the vision sensor	A_1, A_2 B_1, B_2 C_1, C_2 D_1, D_2	0.8771, 0.8748 0.134, 0.0044 0.4801, 0.4844 51.63, 71.57

with the design parameter values, the corresponding error values can be obtained.

According to the error calculation method mentioned above, the model theoretical error value for the solution of pipeline groove typical size parameters (groove width, groove depth) by using the local plane fitting method is obtained under different sensor position and posture changes of the C-LSLVS at the typical parameter conditions in Table 2, as shown in Fig. 16.

In addition, the diameter and wall thickness of the pipe are also important factors affecting the theoretical model error value. The error value of the theoretical model under different pipe diameters is shown in Fig. 17.

From Fig. 16, it can be seen that the change of position and posture of the C-LSLVS has a relatively small impact on the theoretical model error, which is basically kept at the order of 10^{-3} mm. The reason for the change of this error curve is that the groove range corresponding to the laser lines projected on the pipe outer surfaces and its groove surfaces changes with the difference of the position and posture of the vision sensor; the closer the direction of the projected laser lines of the vision sensor is to the radial direction of the pipe, the smaller the groove range corresponding to the projected laser lines is, and the smaller the theoretical model error value is. From Fig. 17, it can be seen that as the curvature radius of the pipe decreases, the curvature of the pipe outer surfaces and its groove surfaces within the groove range corresponding to the two projected laser lines increases, and the error value of the theoretical model increases sharply. When the theoretical model error is relatively small (basically in the order of 10^{-3} mm), it can be ignored compared to the detection accuracy on the order of 10^{-1} mm, which can meet the welding control

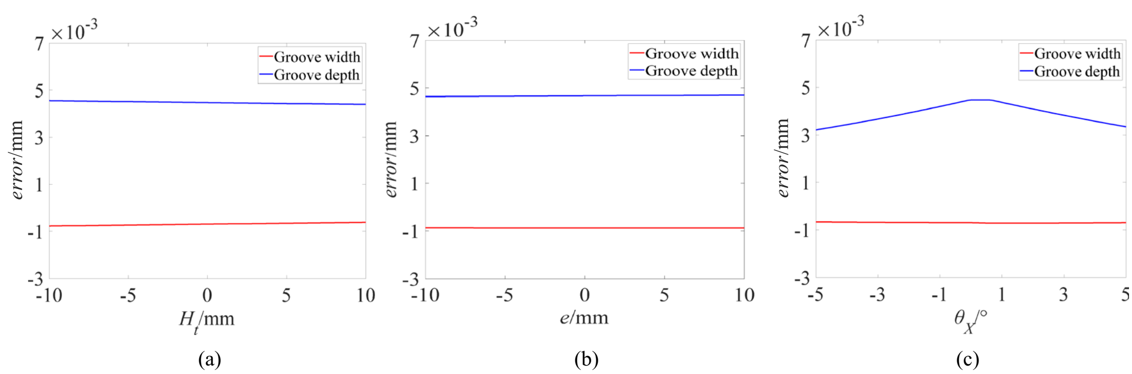


Fig. 16 Theoretical model error for the solution of pipeline groove typical size parameters under different position and posture changes of the C-LSLVS: (**a**) The sensor height (H_t) changes; (**b**) The sensor lateral deviation (e) changes; (**c**) The sensor angle (θ_X) swing changes

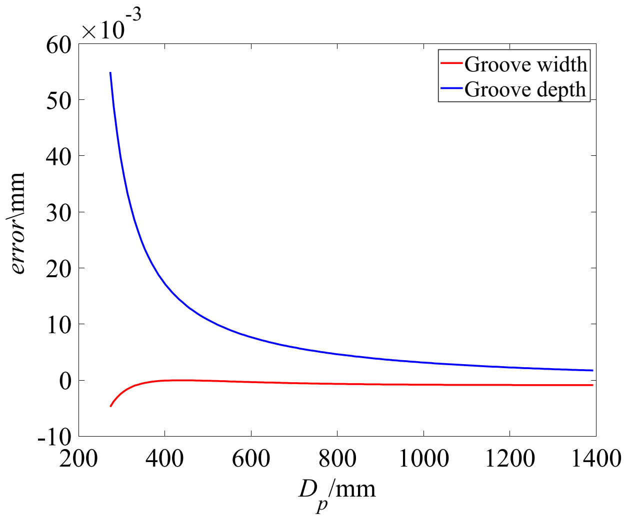


Fig. 17 The theoretical model error for the solution of pipeline groove typical size parameters under different pipe diameters (D_p)

requirements. On the contrary, plane fitting can still be used to replace the local curved surface, but it is necessary to perform corresponding model compensation according to the calculated model error value. Similarly, based on the calculation model established in this section, it can also calculate the theoretical error under other different parameter conditions.

The above establishment and analysis of the model have expanded the application scope of plane fitting method, providing theoretical data support and applicability guidance for solving groove size parameters by using the plane fitting method when the C-LSLVS is applied to the curved workpiece.

5 Experiment and analysis

To realize the pipeline all-position intelligent welding, this paper constructs a pipeline all-position intelligent welding test system based on vision sensing as shown in Fig. 18. The constructed test system mainly consisted of a DW-EW-I pipeline automatic welding robot with single welding torch, an industrial control computer (CPU, AMD Ryzen 5 5600H; GPU, NVIDIA GeForce GTX 1650; RAM, 16G), the designed vision sensor (specific parameters are shown in Table 1 and Table 3), a welding machine controller (ZMC406 controller), a welding power source (MEGMEET Artsen Plus 400), welding shielding gas supply system (20%CO_2 + 80%Ar), and so on.

The pipeline all-position welding adopts the bidirectional downward welding mode, which starts from the 0° position at the top of the pipeline and moves clockwise or counterclockwise to the 180° position at the bottom of the pipeline. The welding method was gas metal arc welding (GMAW). The welding parameters and the required calibration parameters of vision sensor for detection are shown in Table 3. In addition, the relevant designed parameters of pipeline external welding groove are the same as in Table 2.

5.1 Parameters detection test

Before welding, to verify the accuracy of the proposed image processing algorithm of double-V composite groove, and the effectiveness of the calculation and analysis models of welding groove sizes by using plane fitting instead of the local curved surface, the groove size parameters and the lateral deviation between welding torch and groove center were

Fig. 18 The pipeline all-position intelligent welding system based on vision sensing

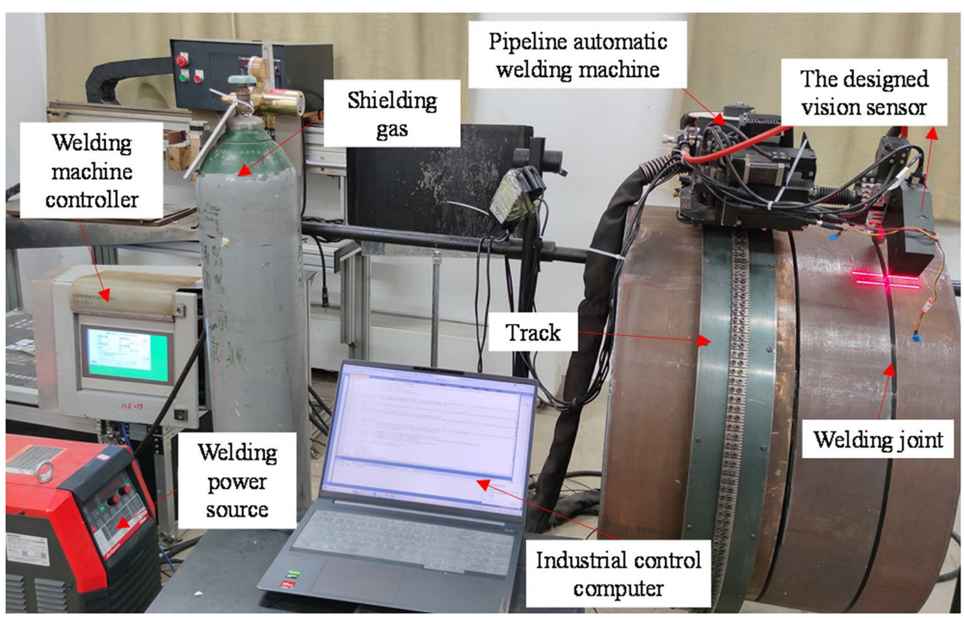


Table 3 Welding parameters and vision sensor calibration parameters

Welding parameters	Value	Sensor calibration parameters	Parameters	Value
Welding wire	1.0 mm	Internal parameters of the camera	f_x, f_y	3544, 35,433
Welding layers	5		u_0, v_0	1239, 1225
Welding passes	6		k_1, k_2	−0.051, 0.074
Welding voltage	25 V	Structured light plane equation parameters of 2 laser emitters	A_1, A_2	0.8771, 0.8748
Welding current	190 A		B_1, B_2	0.134, 0.0044
Welding speed	480 mm/min		C_1, C_2	0.4801, 0.4844
			D_1, D_2	51.63, 71.57

measured using the C-LSLVS at ten different positions of the pipe. Considering the height of the welding torch during arc welding process and the installation height of the sensor, the object distance of the camera is controlled at 140–160 mm in the detection test, which can ensure the similar magnification of the camera in the welding tests. In this case, the resolution of the camera is about 0.03–0.04 (mm/pixel) in the width direction and 0.07–0.10 (mm/pixel) in the depth direction. The measurement data are shown in Table 4.

From Table 4, the maximum absolute error of welding groove width and groove depth detected by vision sensing at ten different positions of the pipe did not exceed 0.14 mm and 0.20 mm, respectively; the maximum absolute error of the lateral deviation detection value between welding torch and groove center did not exceed 0.11 mm (Bold values indicate the maximum absolute error in the column). These demonstrate the practicability of applying the image processing algorithm and the calculation model in this paper to achieve pipeline all-position intelligent welding.

5.2 Image processing test

During the pipeline intelligent welding experiment, in order to verify the anti-interference and stability of the PTFS image processing algorithm, the algorithm and the traditional space domain image processing (SDIP) method [20] were compared under different image conditions. Figure 19 shows the results of two image processing algorithms for images under different conditions.

The ten groups of images in Fig. 19(a) and (b) were respectively collected before and after welding and groove surface polishing during the pipeline all-position welding process. Therefore, the laser line image in Fig. 19(a) has less interference, while the laser line image in Fig. 19(b) has more interference. It can be seen from the comparison in Fig. 19 that the PTFS image processing algorithm maintains a success rate (the proportion of images processed successfully in a group of images) of more than 90% in both cases, while the traditional SDIP method has poor adaptability and anti-interference ability when the images have more interference. Under the system composition in this paper, the proposed algorithm took an average of about 100 ms to perform feedback control of image processing, which was fully satisfactory for the pipeline intelligent welding control at a welding speed of 480 mm/min (or even 600 mm/min). Furthermore, when a dedicated industrial computer is used in the system, less time consumption can be achieved.

Table 4 Pipe welding groove size parameters and welding torch relative lateral deviation detection

Detection position	Detection parameter name								
	Groove width (B/mm)			Groove depth (h/mm)			Welding torch relative lateral deviation (e/mm)		
	Measured value	Detection value	Absolute error	Measured value	Detection value	Absolute error	Measured value	Detection value	Absolute error
0°	8.08	8.22	**0.14 (max)**	20.35	20.19	0.16	0.02	0.06	0.04
20°	8.25	8.31	0.06	20.28	20.14	0.14	0.38	0.33	0.05
40°	8.20	8.29	0.09	20.18	20.14	0.04	0.45	0.46	0.01
60°	8.15	8.27	0.12	20.33	20.21	0.12	0.70	0.79	0.09
80°	8.23	8.29	0.06	20.42	20.36	0.06	1.09	0.98	**0.11 (max)**
100°	8.16	8.27	0.11	20.31	20.20	0.11	1.60	1.66	0.06
120°	8.20	8.26	0.06	20.25	20.16	0.09	1.77	1.69	0.08
140°	8.13	8.23	0.10	20.38	20.45	0.07	2.25	2.19	0.06
160°	8.22	8.30	0.08	20.31	20.11	**0.20 (max)**	2.60	2.57	0.03
180°	8.26	8.33	0.07	20.42	20.37	0.05	3.08	2.98	0.10

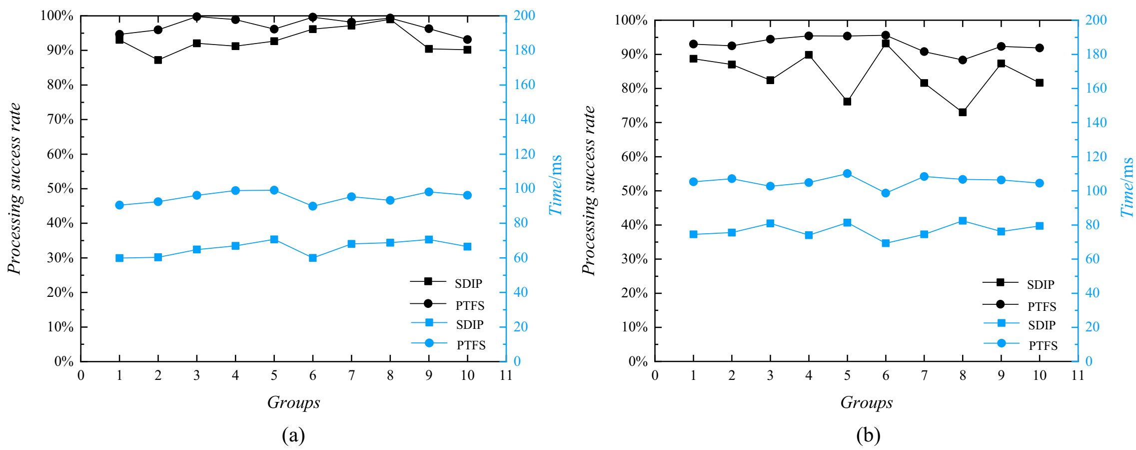


Fig. 19 Comparison of image processing results between SDIP and PTFS under different conditions: Deformed laser line image group without interferences (**a**) and with interferences (**b**)

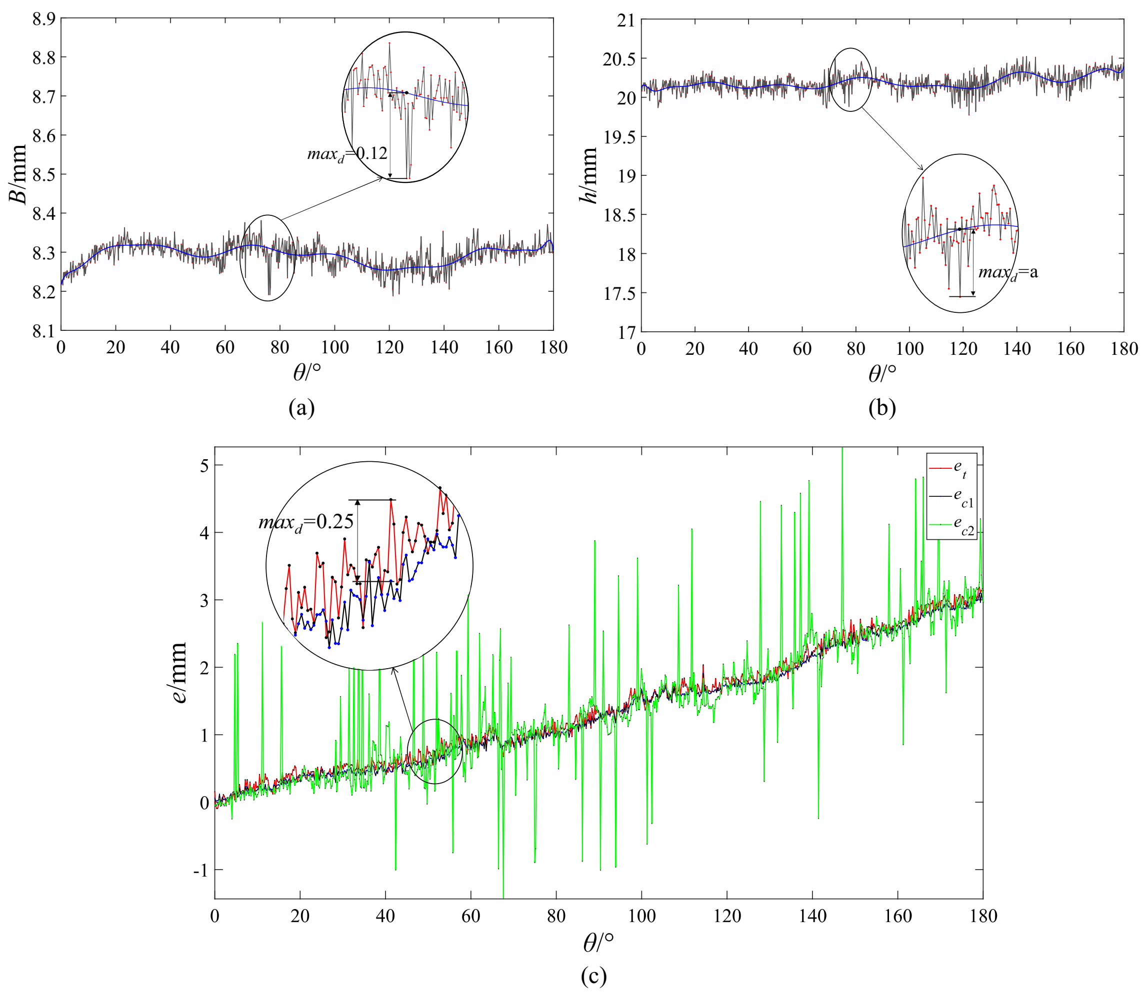


Fig. 20 Parameters detection and welding torch tracking data of the pipeline all-position welding process: (**a**) Groove width detection value; (**b**) Groove depth detection value; (**c**) The welding torch relative lateral deviation and tracking data

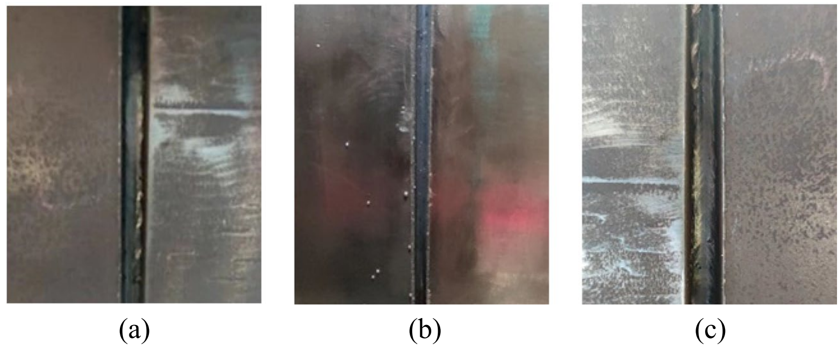


Fig. 21 Typical positions weld forming effect diagram after the pipeline intelligent welding: (**a**) 0° position; (**b**) 90° position; (**c**) 180° position

5.3 Pipeline intelligent welding test

Since the track installation for pipeline welding equipment cannot guarantee strictly parallelism with the groove direction, and there will always be deviations in groove machining and assembly, it is necessary to detect the welding groove (weld bead) size parameters and welding torch relative position deviation during pipeline welding process, so as to adjust the action of the welding torch. Figure 20 shows the groove size parameters obtained by the method proposed in this paper during hot welding, as well as the relative lateral deviation and tracking data of the welding torch obtained by two different image processing algorithms. Figure 21 shows the forming effect of weld bead at different positions after pipeline all-position intelligent welding based on visual servo control.

In Fig. 20, the horizontal axis is the position of the pipe, and the vertical axes of Fig. 20(a) and (b) are the visual detection values of groove width and depth, respectively. The central curve in the figure is a spline curve fitted based on the detection data points. The maximum discrete differences of the data points in Fig. 20(a) and (b) were 0.12 mm and 0.35 mm, respectively. In Fig. 20(c), the e_t in the vertical axis represents the lateral deviation between welding torch and groove center detected by manual teaching, and e_{c1} and e_{c2} respectively represent the welding torch deviation detection value calculated by PTFS image processing algorithm and SDIP method during the welding seam tracking process. From Fig. 20(c), the detected values e_{c2} were unstable and seriously affected by interferences. While the maximum difference between e_t and e_{c1} was 0.25 mm, which verifies the robustness of the proposed image processing method and accuracy of calculation and analysis model, thereby achieving the intelligent control of the welding torch during the pipeline all-position welding process.

From Fig. 21, the weld formation position was accurate, and the formation effect was good, providing strong practical support for the promotion and application of pipeline intelligent welding production.

6 Conclusions

Based on the C-LSLVS, the pipeline all-position intelligent welding is realized, and the following are the main conclusions obtained:

(1) A PTFS image processing algorithm is proposed to extract the laser centerline for the deformed laser lines image of double-V composite groove during the pipeline external welding process, which overcomes the problems of unclear and uneven laser line features, random interference, and polished reflective interference in the images. The algorithm first utilizes information in the time domain to remove random interference from the image. Subsequently, based on RT and DFT, other stationary interferences can be eliminated and laser lines characteristics can be enhanced by back-projection reconstruction in the frequency domain and space domain, thereby achieving stable and accurate extraction of laser centerline under different situations. This lays the credible foundation for subsequent high-precision parameter calculations.

(2) The parameters calculation method based on local plane fitting is applied to the calculation of pipeline groove sizes, and the theoretical analysis and error calculation of the application model are carried out, which provides theoretical data support and adaptive guidance for parameters solving by using the method of plane fitting when the C-LSLVS is applied to the curved workpieces.

(3) A pipeline intelligent welding system based on the C-LSLVS is constructed, and pipeline intelligent welding experiments were carried out. The experiments showed that the groove size detection error was < 0.20 mm, and the weld seam tracking deviation was < 0.25 mm. The results imply that the system can autonomously achieve real-time control of the welding torch and obtain accurate and high-quality weld formation effects, which has important application value for the development of pipeline intelligent welding.

Author contribution Chuanhui Zhu: investigation, methodology, writing — original draft. Zihao Wang: validation, writing — review and editing. Zhiming Zhu: funding acquisition, resources, supervision, writing — review and editing. Jichang Guo: funding acquisition, supervision — review and editing.

Funding The work was supported by the National Natural Science Foundation of China (Grant Nos. 51775301 and 52104162) and Shunyi District "Beijing Scientific and Technological Achievements Transformation Coordination and Service Platform" construction special project.

Declarations

Competing interests The authors declare no competing interests.

References

1. An Y (2017, Jun 10-11) Application of EPC management model in construction of long-distance pipeline project. International Conference on Advances in Materials, Machinery, Electrical Engineering (AMMEE 2017), Tianjin, China: 202–207
2. Hu Y, Lu L, Zhan S (2023) Simulation and reliability evaluation of automated ultrasonic testing technology in semi-automatic welding of oil and gas pipelines. Int J Adv Manuf Tech 124(11):4131–4141
3. Ou Z, Xu C, Han Z, Cen J, Dong S (2024) Nondestructive testing method for internal defects in ferromagnetic materials under weak bias magnetization. Measurement 226:114173
4. Sharma SK, Maheshwari S (2017) A review on welding of high strength oil and gas pipeline steels. J Nat Gas Sci Eng 38:203–217
5. Rout A, Deepak B, Biswal BB, Mahanta GB (2022) Weld seam detection, finding, and setting of process parameters for varying weld gap by the utilization of laser and vision sensor in robotic arc welding. IEEE Trans Ind Electron 69(1):622–632
6. Hao L, Wang H, Shen Y, Wang X, Xu K (2023) A novel multi-seam extraction method for structured workpieces with medium-thick plates based on DLP vision. Measurement 223:113808
7. Wang B, Hu S, Sun L, Freiheit T (2020) Intelligent welding system technologies: state-of-the-art review and perspectives. J Manuf Syst 56:373–391
8. Yang LY, Peng J (2020) Advances techniques of the structured light sensing in intelligent welding robots: a review. Int J Adv Manuf Tech 110(3–4):1027–1046
9. Perez L, Rodriguez I, Rodriguez N, Usamentiaga R, Garcia DF (2016) Robot guidance using machine vision techniques in industrial environments: a comparative review. Sensors 16(3):335
10. Zhao Z, Luo J, Wang Y, Bai L, Han J (2021) Additive seam tracking technology based on laser vision. Int J Adv Manuf Tech 116(1):197–211
11. Huang Y, Xu S, Gao X, Wei C, Zhang Y, Li M (2023) Feature point identification in fillet weld joints using an improved CPDA method. Appl Sci-Basel 13(18):10108
12. Zhang L, Ke W, Ye Q, Jiao J (2014) A novel laser vision sensor for weld line, detection on wall-climbing robot. Opt Laser Technol 60:69–79
13. Li W, Mei F, Hu Z, Gao X, Yu H, Housein AA, Wei C (2022) Multiple weld seam laser vision recognition method based on the IPCE algorithm. Opt Laser Technol 155:108388
14. Hou Z, Xu Y, Xiao R, Chen S (2020) A teaching-free welding method based on laser visual sensing system in robotic GMAW. Int J Adv Manuf Tech 109(5–6):1755–1774
15. Fan J, Jing F, Yang L, Long T, Tan M (2019) A precise seam tracking method for narrow butt seams based on structured light vision sensor. Opt Laser Technol 109:616–626
16. Gu W, Xiong Z, Wan W (2013) Autonomous seam acquisition and tracking system for multi-pass welding based on vision sensor. Int J Adv Manuf Tech 69(1–4):451–460
17. Zou Y, Chen X, Gong G, Li J (2018) A seam tracking system based on a laser vision sensor. Measurement 127:489–500
18. Zhang T, Zhu Z, Zhu C, Sun B (2022) Visual sensing image processing and feature information extraction for arc welding (in Chinese). J Tsinghua Univ (Sci Technol) 62(1):156–162
19. Liu C, Bhole SD (2013) Challenges and developments in pipeline weldability and mechanical properties. Sci Technol Weld Join 18(2):169–181
20. Zhu Z, Zhu C (2023) Laser line image feature extraction system and method for double-V composite groove. CN116673597B. China. 2023-10-13.
21. Xu F, Hou Z, Xiao R, Xu Y, Wang Q, Zhang H (2023) A novel welding path generation method for robotic multi-layer multi-pass welding based on weld seam feature point. Measurement 216:112910
22. Yang L, Fan J, Huo B, Li E, Liu Y (2022) Image denoising of seam images with deep learning for laser vision seam tracking. IEEE Sens J 22(6):6098–6107
23. Wang H, Chi Y, Zhao X, Liu Y, Liang Y, Yang B (2024) Versatile robotic welding system integrating laser positioning, trajectory fitting and real-time tracking. Opt Laser Technol 170:110250
24. Zou Y, Zeng G (2023) Light-weight segmentation network based on SOLOv2 for weld seam feature extraction. Measurement 2023:112492
25. Guo J, Zhu Z, Sun B, Yu Y (2019) Principle of an innovative visual sensor based on combined laser structured lights and its experimental verification. Opt Laser Technol 111(S1):35–44
26. Guo J, Zhu Z, Sun B, Yu Y (2018) A novel multifunctional visual sensor based on combined laser structured lights and its anti-jamming detection algorithms. Weld World 63(2):313–322
27. Zhu C, Zhu Z, Ke Z, Zhang T (2022) Internal parameters calibration of vision sensor and application of high precision integrated detection in intelligent welding based on plane fitting. Sensors 22(6):2117
28. Shao W, Huang Y, Zhang Y (2018) A novel weld seam detection method for space weld seam of narrow butt joint in laser welding. Opt Laser Technol 99:39–51
29. Xue B, Chang B, Peng G, Gao Y, Tian Z, Du D, Wang G (2019) A vision based detection method for narrow butt joints and a robotic seam tracking system. Sensors 19(5):1144
30. Zhu C, Zhu Z, Ke Z, Zhang T (2022) Integrated calibration of internal visual sensor parameters based on combined laser structured lights (in Chinese). J Tsinghua Univ (Sci Technol) 62(9):1516–1523
31. Zeng J, Chang B, Du D, Wang L, Chang S, Peng G, Wang W (2018) A weld position recognition method based on directional and structured light information fusion in multi-layer/multi-pass welding. Sensors 18(1):129
32. Yang J, Zhang L, Li P (2019) Radon-Fourier descriptor for invariant pattern recognition. Int J Wavelets Multiresolut Inf Process 17(2):1940004
33. Zhang H, Wang M, Cheng Z, Jiang K, Zhang M (2020) Filtered back-projection reconstruction algorithm based on new hybrid filter function (in Chinese). Autom Instrum 35(11):94–97
34. Chen W, Sui L, Xu Z, Lang Y (2012, May 19-20) Improved Zhang-Suen thinning algorithm in binary line drawing applications. 2012 International Conference on Systems and Informatics (ICSAI 2012), Yantai, China: 1947–1950